2.66inch e-Paper Module (G) Manual

2.66inch e-Paper raw panel



2.66inch e-Paper Module (G)



360 × 184
RPI, SPI
Black White Yellow Red

Overview

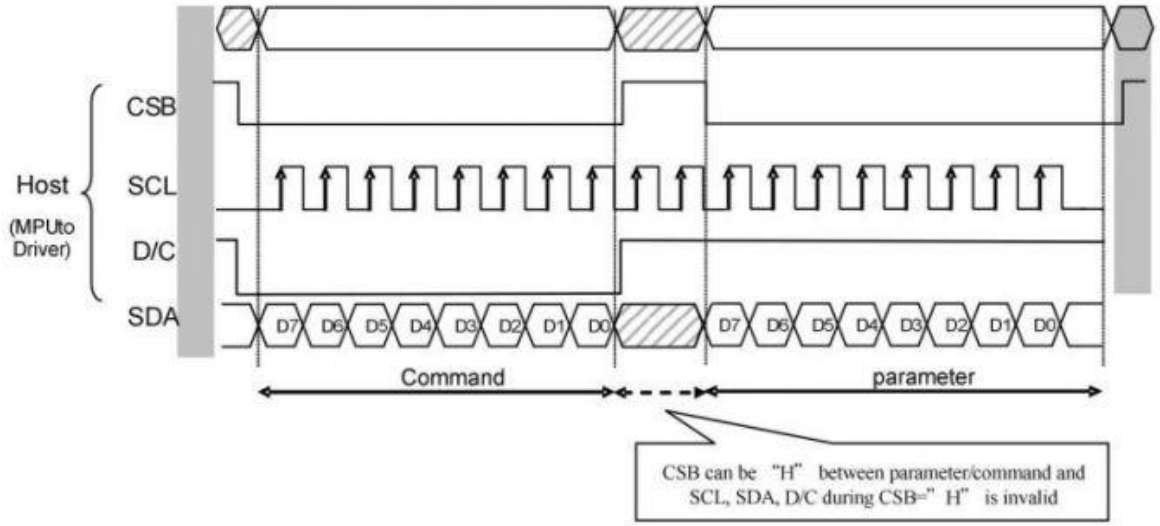
Parameters

Dimensions	2.66inch
Driver board dimensions	85mm × 38.5mm
Display dimensions	60.05mm × 30.69mm
Outline dimensions (screen only)	72.59 × 37.11 × 1mm

Operating voltage	3.3V / 5V (IO level should be consistent with the power supply)
Communication interface	SPI
Dot pitch	0.1668 × 0.1668mm
Resolution	360 × 184
Display color	Black, White, Red, Yellow
Grey scale	2
Refresh time	26s
Refresh power	<50mW (typ.)
Standby current	< 0.01uA (almost 0)
Operating temperature	0 ~ 40 ℃
Storage temperature	-25 ~ 60 ℃

- Refresh time: The refresh time is the experimental results, the actual refresh time will have errors, and the actual effect shall prevail. There will be a flickering effect during the global refresh process, this is a normal phenomenon.
- Refresh power consumption: The power consumption data is the experimental results. The actual power consumption will have a certain error due to the existence of the driver board and the actual use situation. The actual effect shall prevail.
- Refresh in a low temperature environment may appear color cast, it need to be static in the environment of 25℃ for 6 hours before refresh.

Communication Method



CSB (CS): Slave chip select signal, active at low level. When it is at low level, the chip is enabled.
SCL (SCK/SCLK): Serial clock signal.
D/C (DC): Data/command control signal, write command (Command) when the level is low; write data (Data/parameter) when the level is high.
SDA (DIN): Serial data signal.
Timing: CPHL=0, CPOL=0, i.e. SPI mode 0.
Note: For specific information about SPI communication, you can search for information online on your own.

Operating Principle

The e-paper used in this product uses "microcapsule electrophoresis display" technology for image display. The basic principle is that charged nanoparticles suspended in a liquid migrate under the action of an electric field. The e-paper display screen displays patterns by reflecting ambient light and does not require a backlight. Under ambient light, the e-paper display screen is clearly visible, with a viewing angle of almost 180°. Therefore, e-paper displays are ideal for reading.

Program Principle

For a four-color e-Paper screen, we can refer to black and white four-level grayscale images for understanding, but there are still some differences. To save memory space, the e-Paper screen compresses pixels as follows:
Black: Corresponds to 00b
White: Corresponds to 01b
Yellow: Corresponds to 10b
Red: Corresponds to 11b
Let's take four pixels as an example:

pixel	1	2	3	4

bit	0	1	2	3	4	5	6	7
Stored Data	0	0	0	1	1	0	1	1
Color	Black		White		Yellow		Red	
byte	0x1B							

Writing the data 0x1B to register 0x10 will set the four pixels on the e-Paper screen to display black, white, yellow, and red colors based on the encoding scheme mentioned earlier.

Precautions

1. For e-Paper displays that support partial refresh, please note that you cannot refresh them with the partial refresh mode all the time. After refreshing partially several times, you need to fully refresh EPD once. Otherwise, the display effect will be abnormal.
2. It is a normal phenomenon that the three-color or multi-color EPD will have a certain color difference in different batches. Hence, it is recommended to use the program to clear all the pictures on the EPD and store it facing up. Please clear the screen several times before powering on.
3. Note that the screen cannot be powered on for a long time. When the screen is not refreshed, please set the screen to sleep mode or power off it. Otherwise, the screen will remain in a high voltage state for a long time, which will damage the e-Paper and cannot be repaired!
4. When using the e-Paper display, it is recommended that the refresh interval be at least 180s, and refresh at least once every 24 hours. If the e-Paper is not used for a long time, you should use the program to clear the screen before storing it. (Refer to the datasheet for specific storage environment requirements.)
5. After the screen enters sleep mode, the sent image data will be ignored, and it can be refreshed normally only after initializing again.
6. Control the 0x3C or 0x50 (refer to the datasheet for details) register to adjust the border color. In the demo, you can adjust the Border Waveform Control register or VCOM AND DATA INTERVAL SETTING to set the border.
7. If you find that the created image data is displayed incorrectly on the screen, it is recommended to check whether the image size setting is correct, change the width and height settings of the image and try again.
8. The working voltage of the e-Paper display is 3.3V. If you buy the raw panel, you need to add a level convert circuit for compatibility with 5V voltage. The new version of the driver board (V2.1 and subsequent versions) has added a level

processing circuit, which can support both 3.3V and 5V. The old version only supports a 3.3V working environment. You can confirm the version before using it. (The version number is under the board name.)

9. The FPC cable of the screen is fragile, Please note: Do not bend the cable along the vertical direction of the screen to avoid tearing the cable; Do not repeatedly excessive bending the cable, to avoid the cable fracture; Do not bend the cable toward the front of the screen to avoid the cable from being disconnected from the panel. It is recommended to use fixed wiring during debugging and development.
10. The screen of e-Paper is relatively fragile, please try to avoid dropping, bumping and pressing hard.
11. We recommend that customers use the sample program provided by us to test with the corresponding development board.

Picture Processing

Picture production and conversion of multi-color E-ink.

Picture Production

Preparation

Required software: Adobe PhotoShop CC, Painting

Introduction

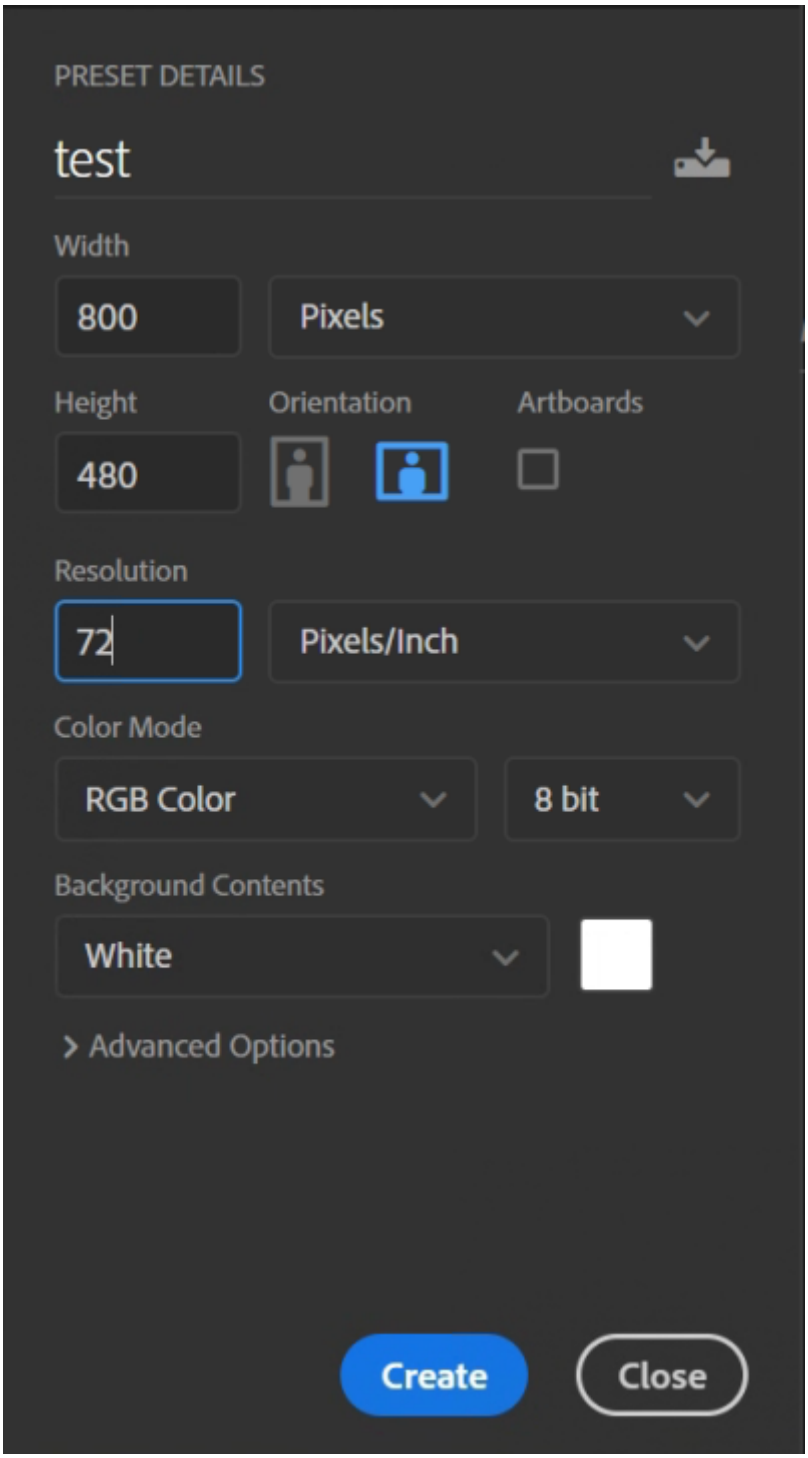
- Floyd-Steinberg dithering algorithm is very suitable for showing rich layering where the number of colors is small. It makes it possible to get more color combinations for better shadow rendering of the original image. It is especially suitable for various usage scenarios of e-ink screens.
- The following will introduce how to convert ordinary pictures into Floyd-Steinberg scatter plots.
- If you are interested in the actual algorithm, you can read about our algorithm porting on ESP32 and ESP8266. Here, no further elaboration will be given.

Operating Steps

Preparation: Download color sheet to PC, and unzip to get the file as shown below, we are going to use "N-color.act" or "4-color.act".



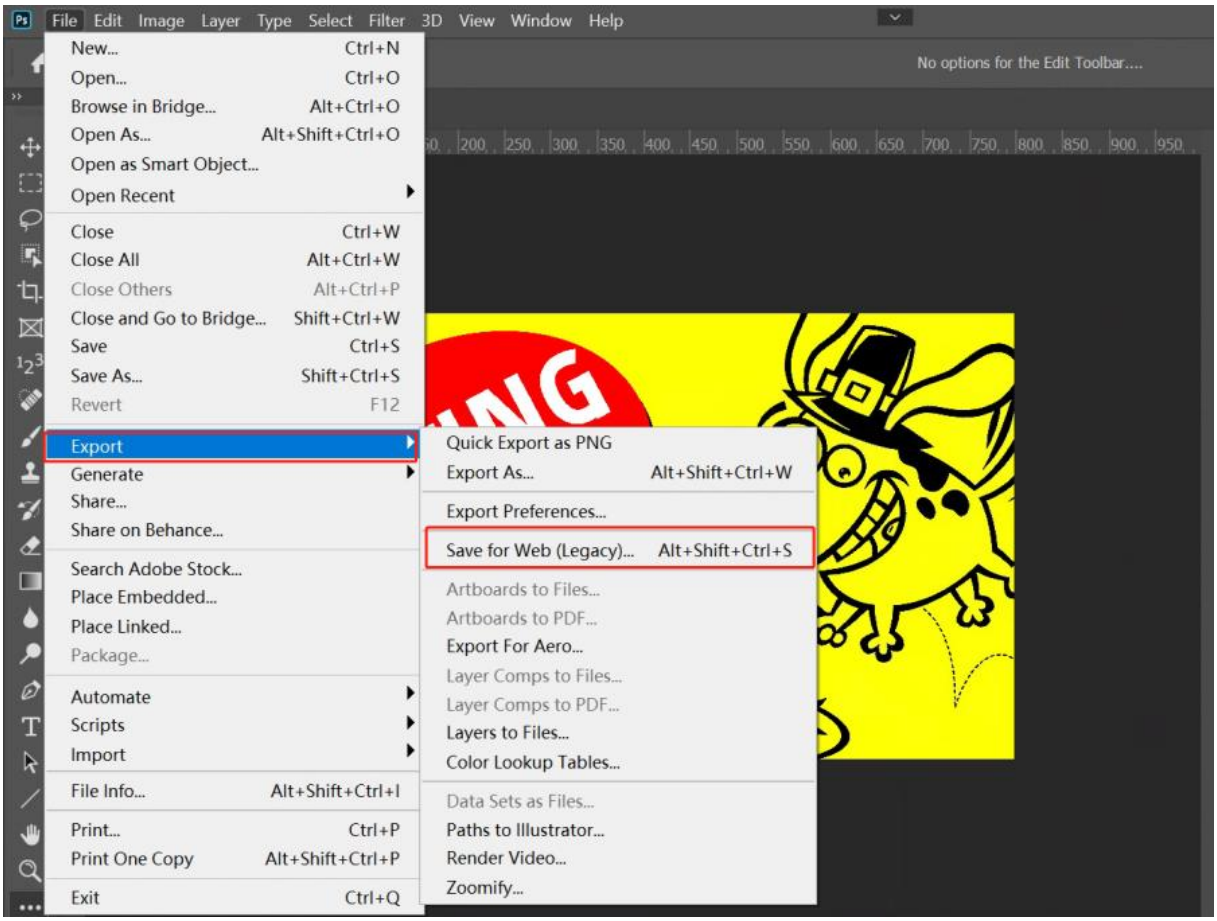
1. The new Photoshop project sets the width and height according to the actual resolution of the e-Paper display, the color mode uses RGB. The resolution of this module is 800*480. Change the width to 800 pixels and the height to 480 pixels.



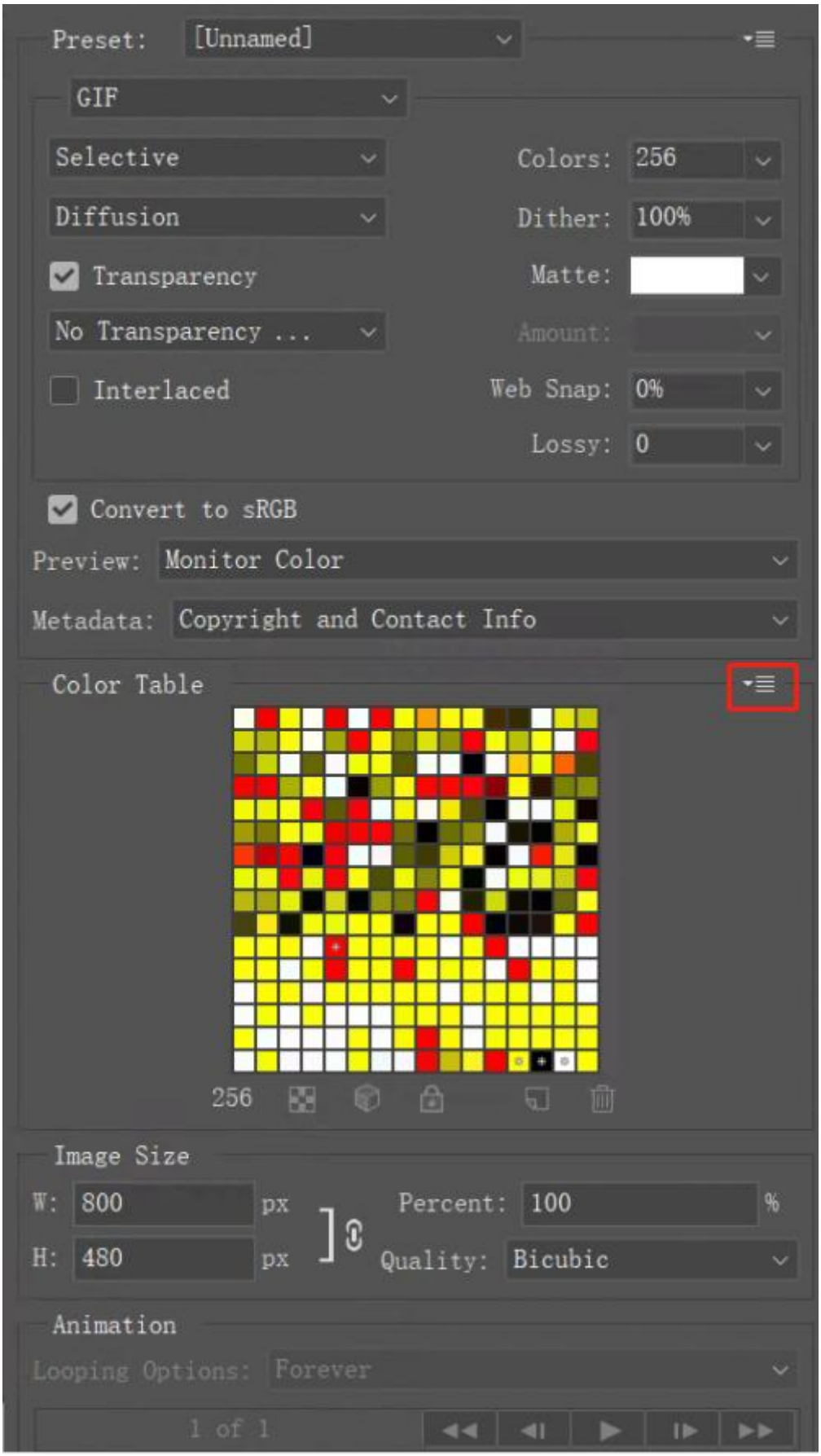
2. Prepare the corresponding picture, copy it into the project, and adjust the parameters such as size and contrast (similar to the steps for processing pictures in general Photoshop).



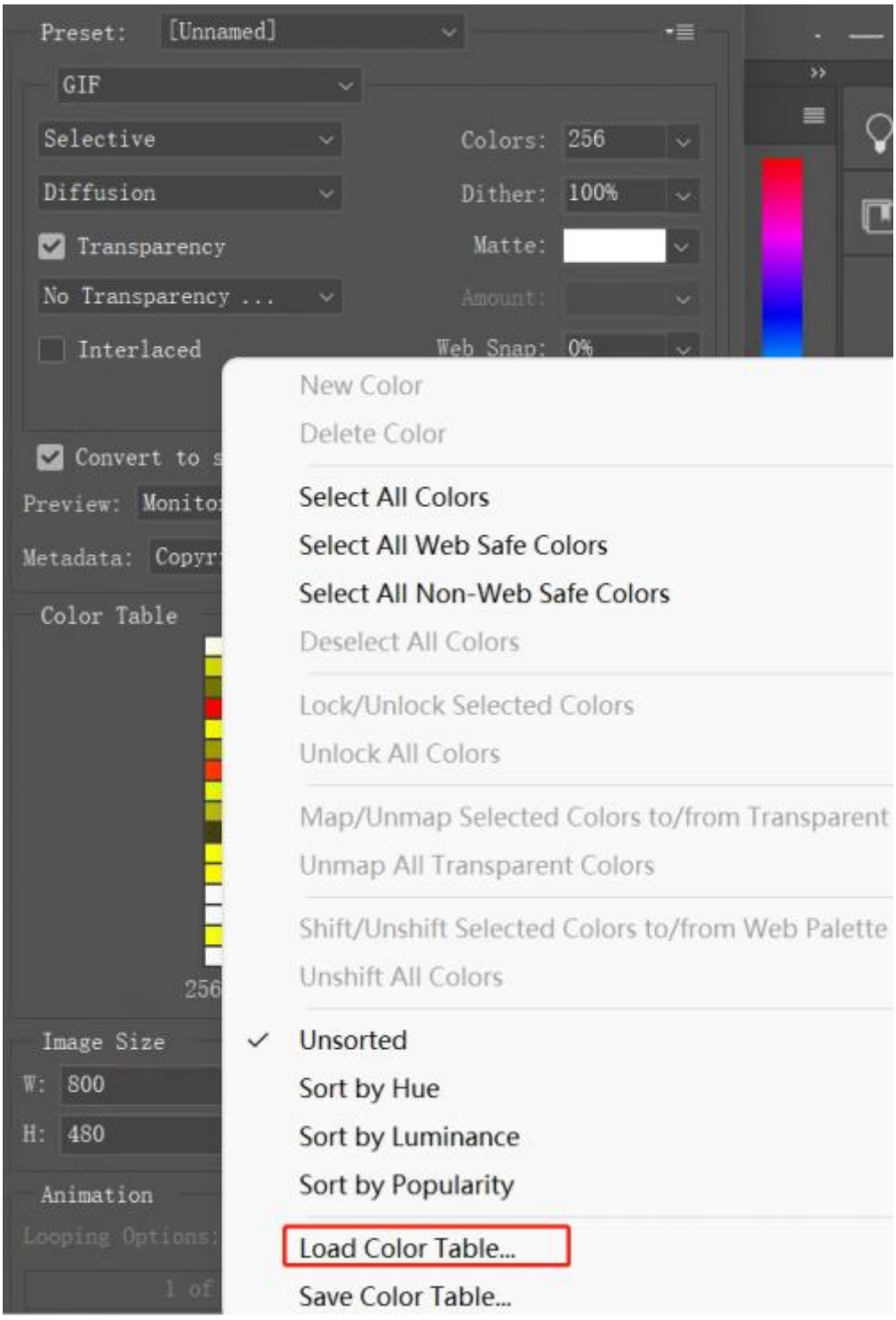
3. Select File -> Export -> Save for Web (Legacy).



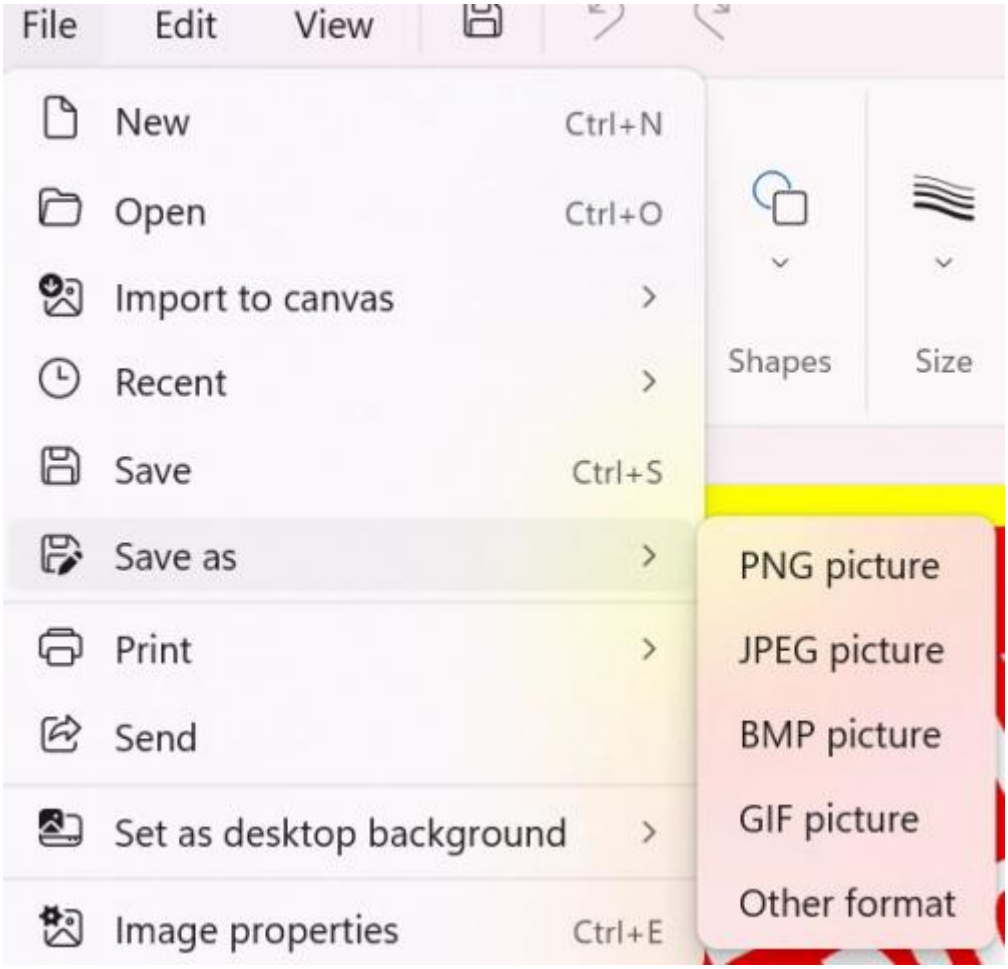
4. Select the color table as shown below. Load the color table provided in Preparation.



5. For seven-color pictures, load N-color. act, then click Save and save as a gif file. Then convert it into BMP format and use it on this module.
For four-color images, load 4-color. act, then click Save and save as a gif file. Then convert it into BMP format and use it on this module.
For six-color images, load 6-color. act, then click Save and save as a gif file. Then convert it into BMP format and use it on this module.



6. Open the gif file with Paint and save it as a 24-bit BMP image.



7. By now, the color picture available for this module has been made, and it can be put into the TF card of the Raspberry Pi or e-Paper Shield module for use, or refer to the next section to convert to an array for use by other embedded devices.

Image Data Conversion

Download the Program

- Four color image conversion application: Coverterto4color
- Six color image conversion application: ConverterTo6color
- Seven color image conversion application: Converterto7color
- Source code: Convertertocolor

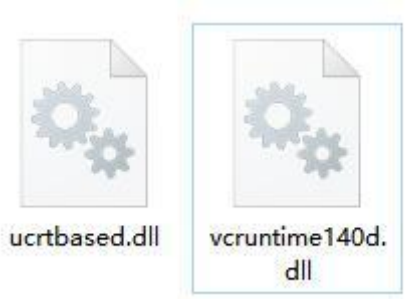
Note: This application is provided for your convenience and open source. Our company does not provide technical support for it.

Bug Fixing

If your computer is running on the Win10/Win11 system and doesn't have Microsoft Visual Studio or any other Microsoft development tools installed, you may encounter

error prompts: "ucrtbased.dll" and "vcruntime 140d.dll".
This indicates that your computer lacks these two components. Here's the solution:

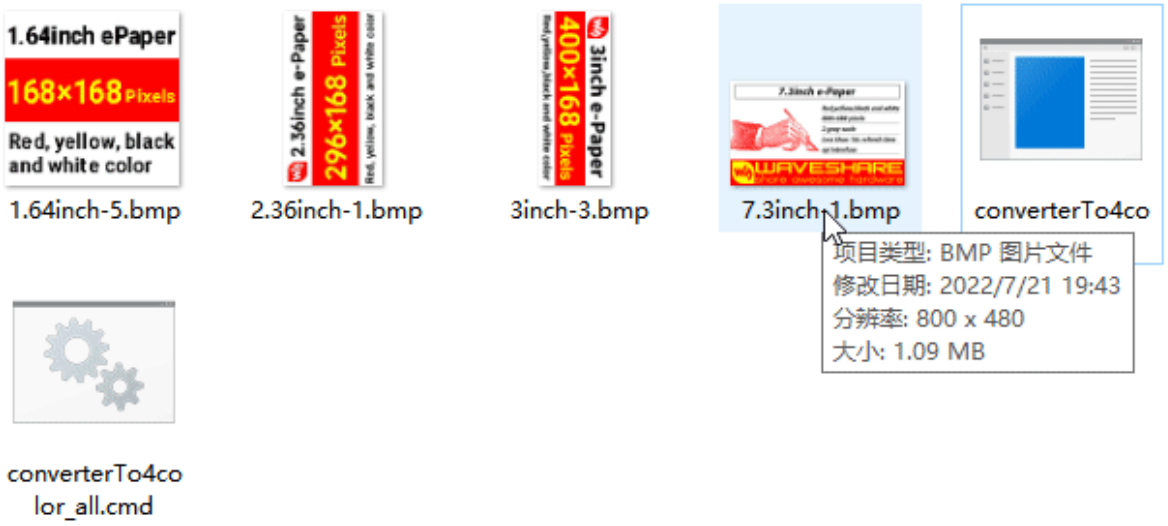
1. Install VS (Microsoft Visual Studio) or other Microsoft development tools (such as Visual C++ Redistributable).
2. Use the file we provide:



Place these two files into the directory C:\Windows\System32, then restart your computer.

How to Convert

- Put the prepared picture and the corresponding "exe" application in the same folder, and you can put multiple pictures at the same time.
- Drag and drop the picture in the "exe" file, and the program will convert the picture into a ".c" file with a fixed name.
- Double-click the "cmd" file, and the program will convert all the pictures in the folder that meet the size to generate ".c" files with the corresponding names.
- Click on the image to see the demonstration.
    - Four-color single-image demonstration:



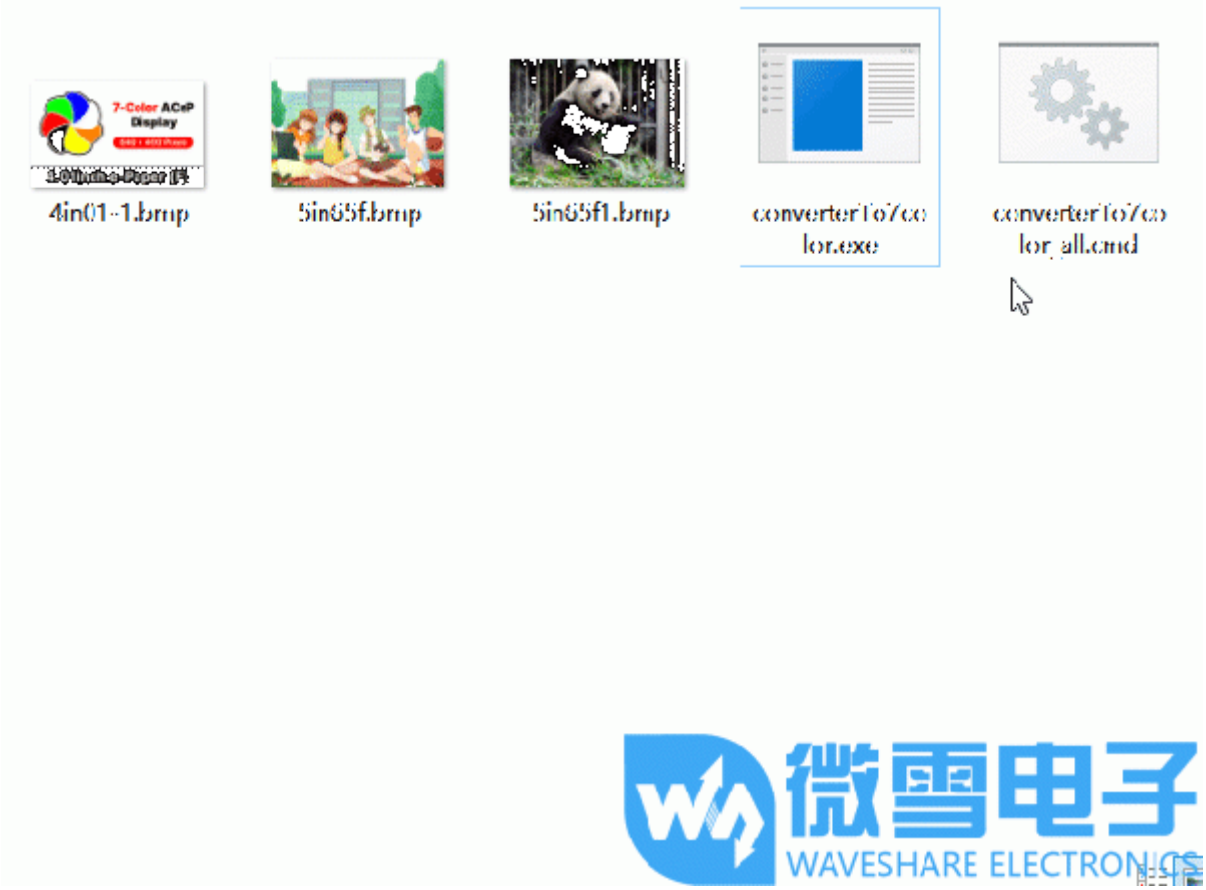
-
    - Four-color multi-picture demonstration:



-
    - Seven-color single-picture demonstration:



-
    - Seven-color multi-picture demonstration:



Working With Raspberry Pi

Hardware Connection

When connecting the Raspberry Pi, you can directly insert the board into the 40PIN pin header of the Raspberry Pi, and pay attention to the correct pins.
If you choose to connect with an 8PIN cable, please refer to the pin correspondence table below:

Raspberry Pi Pinout

e-Paper	Raspberry Pi BCM2835	Board
VCC	3.3V	3.3V
GND	GND	GND
DIN	MOSI	19
CLK	SCLK	23
CS	CE0	24
DC	25	22

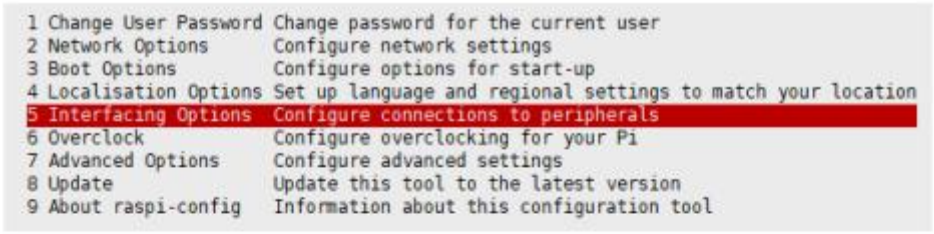
RST	17	11
BUSY	24	18
PWR	18	12

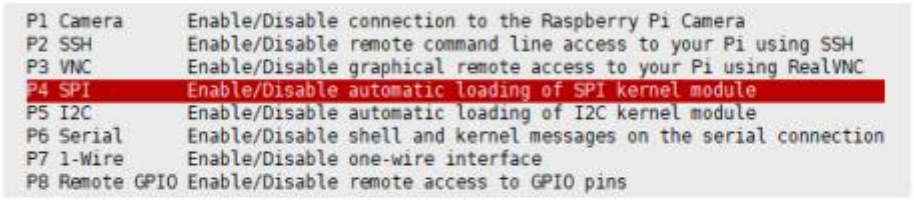
Enable SPI Interface

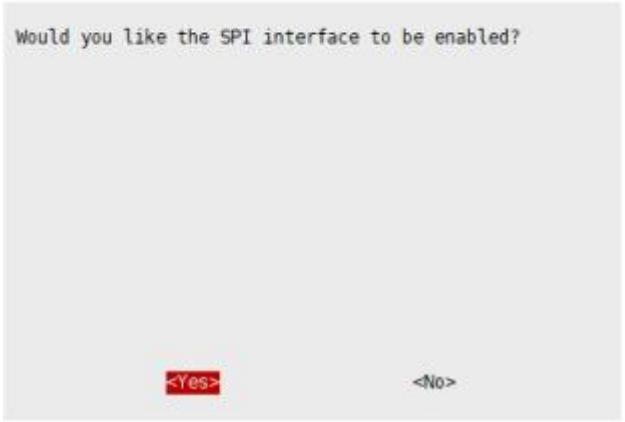
- Open the Raspberry Pi terminal and enter the following command in the config interface:

sudo raspi-config

Choose Interfacing Options -> SPI -> Yes Enable SPI interface



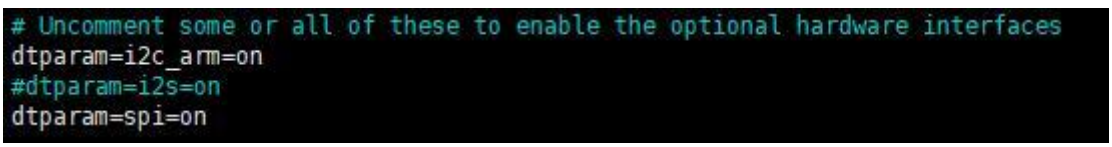




Then reboot your Raspberry Pi:

sudo reboot

- Check /boot/config.txt, and you can see 'dtparam=spi=on' was written in.



- To make sure SPI is not occupied, it is recommended to close other drivers' coverage. You can use ls /dev/spi* to check whether SPI is occupied. If the terminal outputs /dev/spidev0.0 and /dev/spidev0.1, SPI is not occupied.

```
pi@raspberrypi:~ $ ls /dev/spi*
/dev/spidev0.0  /dev/spidev0.1
```

C

- Install lg library

#Open the Raspberry Pi terminal and run the following commands:

wget https://github.com/joan2937/lg/archive/master.zip

unzip master.zip

cd lg-master

make

sudo make install

#For more details, you can refer to the source code: https://github.com/gpiozero/lg

- Install gpiod library (Optional)

#Open the Raspberry Pi terminal and run the following commands:

sudo apt-get update

sudo apt install gpiod libgpiod-dev

- Install BCM2835 (Optional):

#Open the Raspberry Pi terminal and run the following command

wget http://www.airspayce.com/mikem/bcm2835/bcm2835-1.71.tar.gz

tar zxvf bcm2835-1.71.tar.gz

cd bcm2835-1.71/

sudo ./configure && sudo make && sudo make check && sudo make install

For more information, please refer to the official website: http://www.airspayce.com/mikem/bcm2835/

- Install WiringPi (Optional):

#Open the Raspberry Pi terminal and run the following command:

sudo apt-get install wiringpi

#For Raspberry Pi systems after May 2019 (earlier than before, you may not need to execute), you may need to upgrade:

wget https://project-downloads.drogon.net/wiringpi-latest.deb

sudo dpkg -i wiringpi-latest.deb

gpio -v

#Run gpio -v and version 2.52 will appear. If it does not appear, the installation is wrong.

#Bullseye branch system use the following command:

git clone https://github.com/WiringPi/WiringPi

cd WiringPi

./build

gpio -v

Run gpio -v and version 2.60 will appear. If it does not appear, it means that there is an installation error.

- Download the demo via GitHub (You can skip this step if you have downloaded it.)

git clone https://github.com/waveshare/e-Paper.git

cd e-Paper/RaspberryPi_JetsonNano/

- Download the demo (You can skip this step if you have downloaded it.)

wget https://files.waveshare.com/upload/7/71/E-Paper_code.zip

unzip E-Paper_code.zip -d e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Alternative decompression methods:

sudo apt-get install p7zip-full

7z x E-Paper_code.zip -O./e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Compile the demo (Note: -j4 is to compile with 4 threads, the numbers can be modified by yourself; EPD=epd2in66 is to specify a macro definition, and epd2in66 corresponds to the test demo in the main function).

Now at e-Paper/RaspberryPi_JetsonNano

cd c

sudo make clean

sudo make -j4 EPD=epd2in66g

- Run the demo

sudo ./epd

Python

- Install the function library:

sudo apt-get update

sudo apt-get install python3-pip

sudo apt-get install python3-pil

sudo apt-get install python3-numpy

sudo pip3 install spidev

- Install function library (python2):

sudo apt-get update

sudo apt-get install python-pip

sudo apt-get install python-pil

sudo apt-get install python-numpy

sudo pip install spidev

- Install gpiozero library (it is installed in the system by default, if not, you can install it by following the commands below)

sudo apt-get update

python3

sudo apt install python3-gpiozero

python2

sudo apt install python-gpiozero

- Download the demo via GitHub (You can skip this step if you have downloaded it.)

git clone https://github.com/waveshare/e-Paper.git

cd e-Paper/RaspberryPi_JetsonNano/

- Download the demo (You can skip this step if you have downloaded it.)

wget https://files.waveshare.com/upload/7/71/E-Paper_code.zip

unzip E-Paper_code.zip -d e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Alternative decompression method:

sudo apt-get install p7zip-full

7z x E-Paper_code.zip -O./e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Run the demo.

Make sure it's in e-Paper/RaspberryPi_JetsonNano/

cd python/examples/

python3 epd_2in66g_test.py

- If it doesn't run, reboot and call the python file again.

Working With Arduino

Hardware Connection

Use an 8PIN cable to connect, please refer to the pin correspondence table below:

Arduino Pinout

e-Paper	Arduino UNO	Mega2560
VCC	5V	5V
GND	GND	GND
DIN	D11	D51
CLK	D13	D52
CS	D10	D10
DC	D9	D9
RST	D8	D8
BUSY	D7	D7
PWR	D6	D6

Install IDE

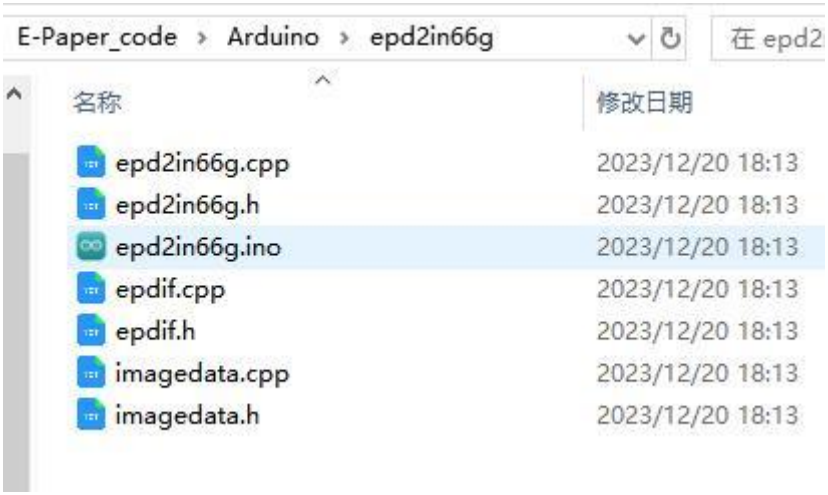
- [Arduino IDE Windows Install Guide]

Run The Demo

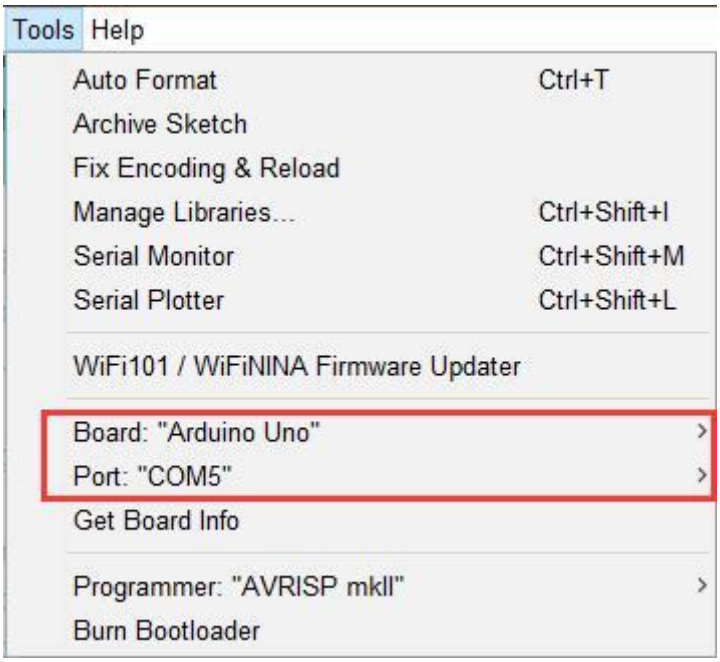
- Download the [demo] in Resource, unzip it to the "E-Paper_code" directory, and you can see the following content:



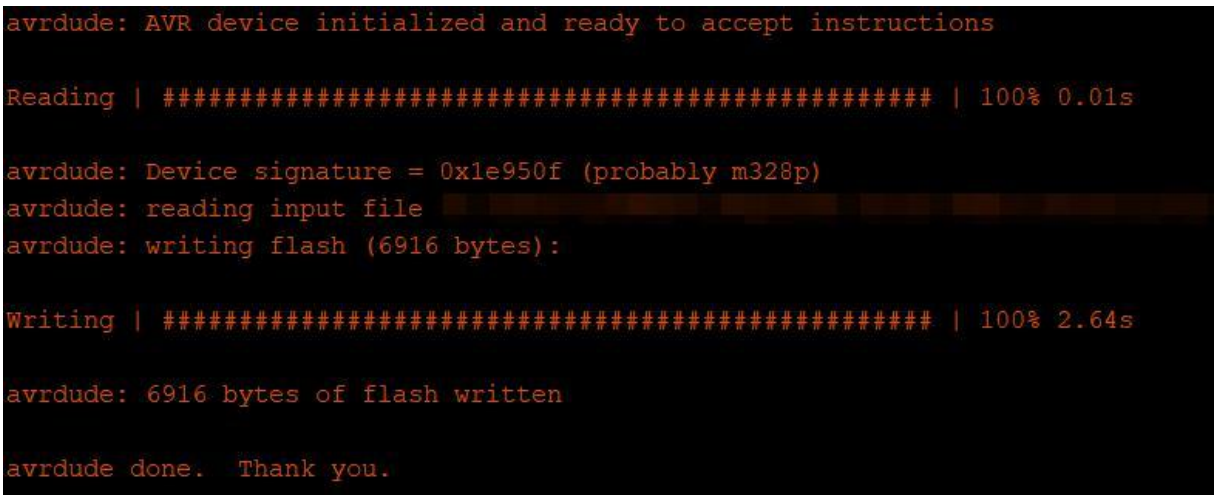
- Open the test demo: E-Paper_code\Arduino\epd2in13g\epd2in66g.ino



- Select the corresponding Board and Port in the Tools in the Arduino IDE.



- Finally, click upload, the upload is successful as follows (Arduino 1.8.13).



Working With Jetson Nano

Hardware Connection

The 40PIN pin of Jetson Nano is compatible with the 40PIN pin of Raspberry Pi and the Jetson.GPIO library with the same API as the RPI.GPIO library of Raspberry Pi is provided, so the serial number connected here is the same as that of Raspberry Pi. The module can be directly inserted into the 40Pin headers of the Jetson Nano when using the 40PIN interface.
If you choose to connect with an 8PIN cable, please refer to the pin correspondence table below:

Jetson nano Pinout

e-Paper	Jetson Nano Developer Kit BCM2835	Board
VCC	3.3V	3.3V
GND	GND	GND
DIN	10 (SPI0_MOSI)	19
CLK	11 (SPI0_SCK)	23
CS	8 (SPI0_CS0)	24
DC	25	22
RST	17	11
BUSY	24	18
PWR	18	12

C

- Download the demo via GitHub (you can skip this step if you have downloaded it.)

```
git clone https://github.com/waveshare/e-Paper.git

cd e-Paper/RaspberryPi_JetsonNano/
```

- Download the test demo: (you can skip this step if you have downloaded it.)

```
wget https://files.waveshare.com/upload/7/71/E-Paper_code.zip

unzip E-Paper_code.zip -d e-Paper

cd e-Paper/RaspberryPi_JetsonNano/
```

- Alternative decompression method:

```
sudo apt-get install p7zip-full

7z x E-Paper_code.zip -O./e-Paper

cd e-Paper/RaspberryPi_JetsonNano/
```

- Compile the demo (Note: JETSON is the specified device, and RPI is not specified by default. -j4 is to compile by 4 threads, and the number can be changed by yourself. "EPD=epd2in66" is to specify a macro definition, and "epd2in66" corresponds to the test demo in the main function.)

```
# Now at e-Paper/RaspberryPi_JetsonNano

cd c

sudo make clean

sudo make JETSON -j4 EPD=epd2in66g
```

- Run the demo

```
sudo ./epd
```

Python

- Install function library:

```
sudo apt-get update

sudo apt-get install python3-numpy

sudo apt-get install python3-pip

sudo pip3 install Jetson.GPIO
```

- Download the demo via GitHub (you can skip this step if you have downloaded it.)

git clone https://github.com/waveshare/e-Paper.git

cd e-Paper/RaspberryPi_JetsonNano/

- Download the demo (you can skip this step if you have downloaded it.)

wget https://files.waveshare.com/upload/7/71/E-Paper_code.zip

unzip E-Paper_code.zip -d e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Alternative decompression method:

sudo apt-get install p7zip-full

7z x E-Paper_code.zip -O./e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Run the demo.

Make sure it's in e-Paper/RaspberryPi_JetsonNano/

cd python/examples/

python3 epd_2in66g_test.py

Working With RDK X3

Hardware Connection

When connecting the RDK X3, you can directly insert the board into the 40PIN pin header of the RDK X3, and pay attention to the correct pins.
If you choose to connect with an 8PIN cable, please refer to the pin correspondence table below:

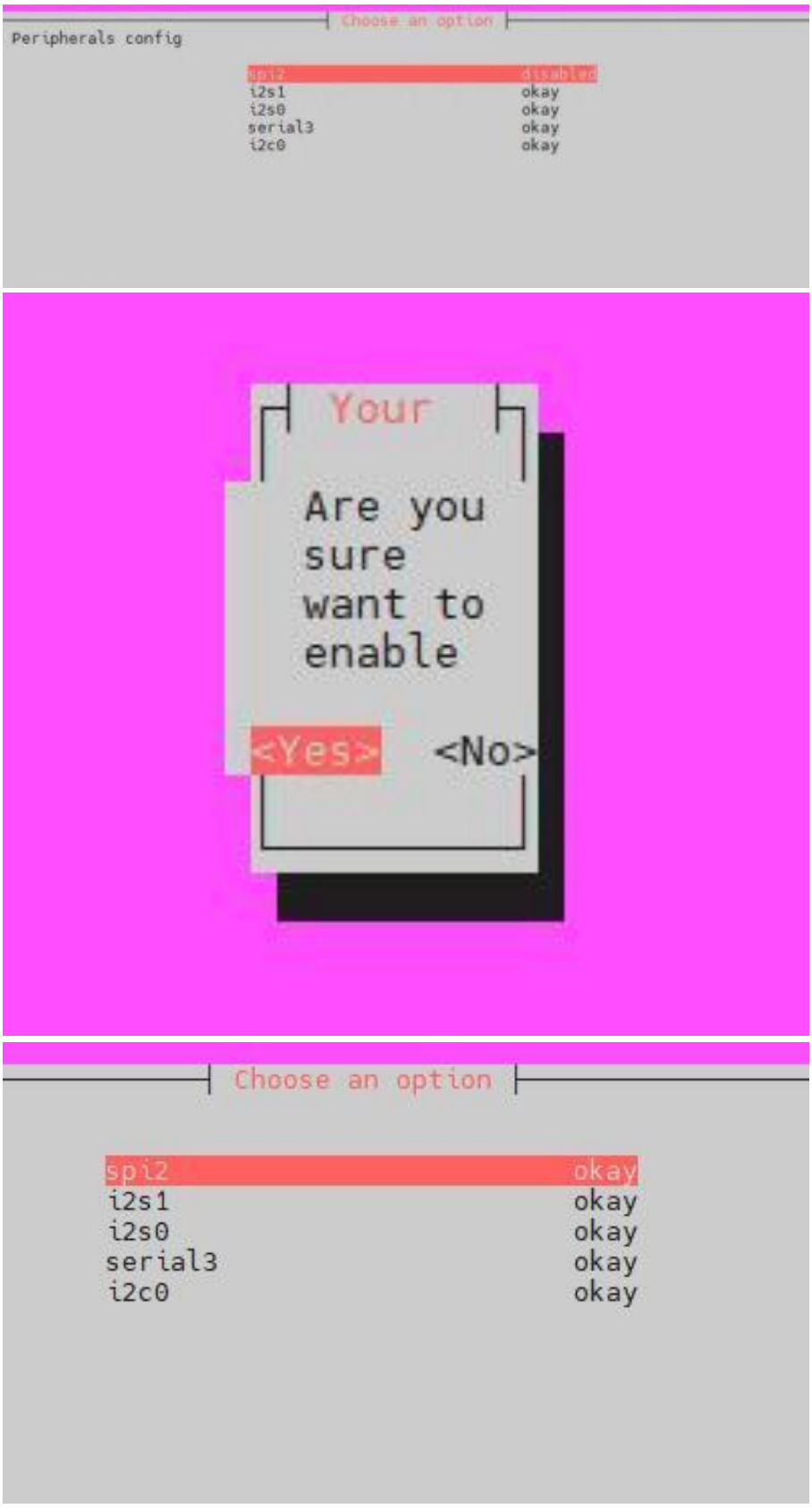
RDK X3 Pinout

e-Paper	RDK X3 BCM	Board
VCC	3.3V	3.3V
GND	GND	GND
DIN	MOSI	19
CLK	SCLK	23
CS	CE0	24

DC	25	22
RST	17	11
BUSY	24	18
PWR	18	12

Enable SPI

- SPI is enabled by default. If you have disabled it, you can enable it by following the steps below.
- Enter the command: sudo srpi-config



Python

- The corresponding library has been installed in the function. If you uninstall it accidentally, please use the following command to install it.

sudo apt-get update

sudo apt-get install python-pip

sudo apt-get install python-pil

sudo apt-get install python-numpy

sudo pip install Hobot.GPIO

sudo pip install spidev

- Download the demo via GitHub (skip this step if you have downloaded it).

git clone https://github.com/waveshare/e-Paper.git

cd e-Paper/RaspberryPi_JetsonNano/

- Download the demo (skip this step if you have downloaded it).

wget https://files.waveshare.com/upload/7/71/E-Paper_code.zip

unzip E-Paper_code.zip -d e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Alternative decompression method:

sudo apt-get install p7zip-full

7z x E-Paper_code.zip -O./e-Paper

cd e-Paper/RaspberryPi_JetsonNano/

- Run the demo.

Make sure you are in e-Paper/RaspberryPi_JetsonNano/

cd python/examples/

python3 epd_2in66g_test.py

Working With STM32

Hardware Connection

If you choose to connect with an 8PIN cable, please refer to the pin correspondence table below:

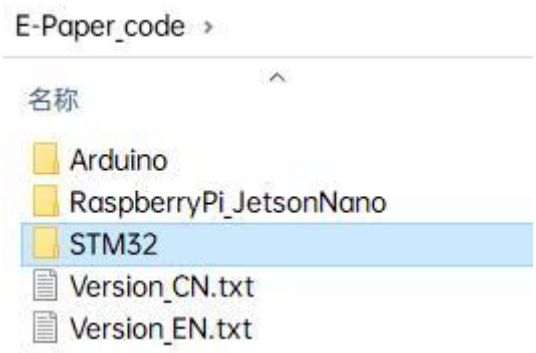
STM32 Pinout

e-Paper	STM32
VCC	3.3V

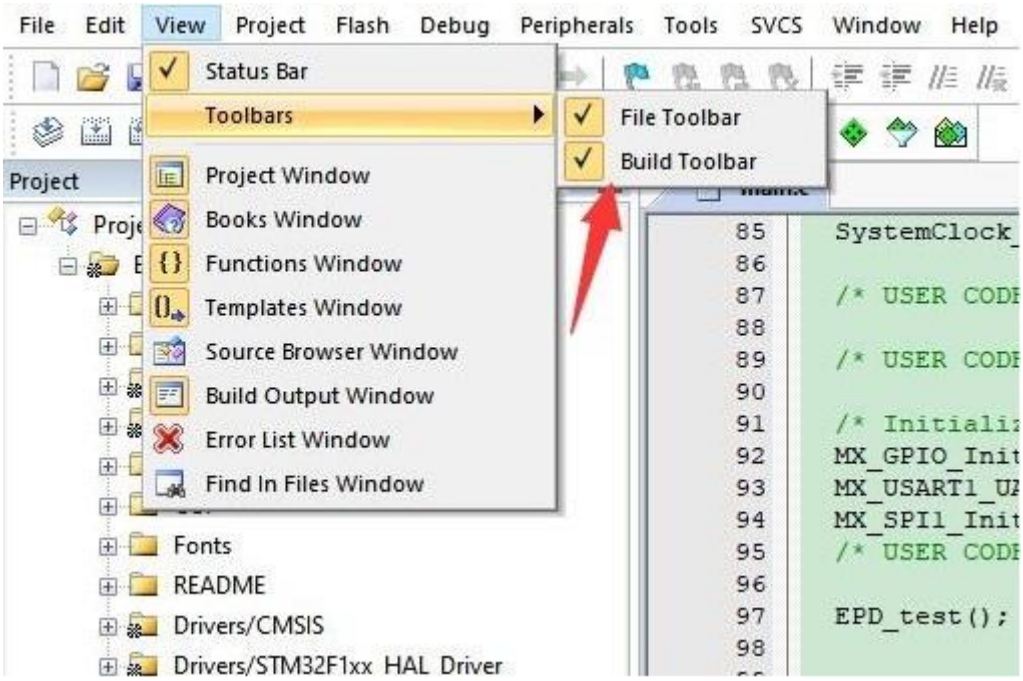
GND	GND
DIN	PA7
CLK	PA5
CS	PA4
DC	PA2
RST	PA1
BUSY	PA3
PWR	PA6

Run The Demo

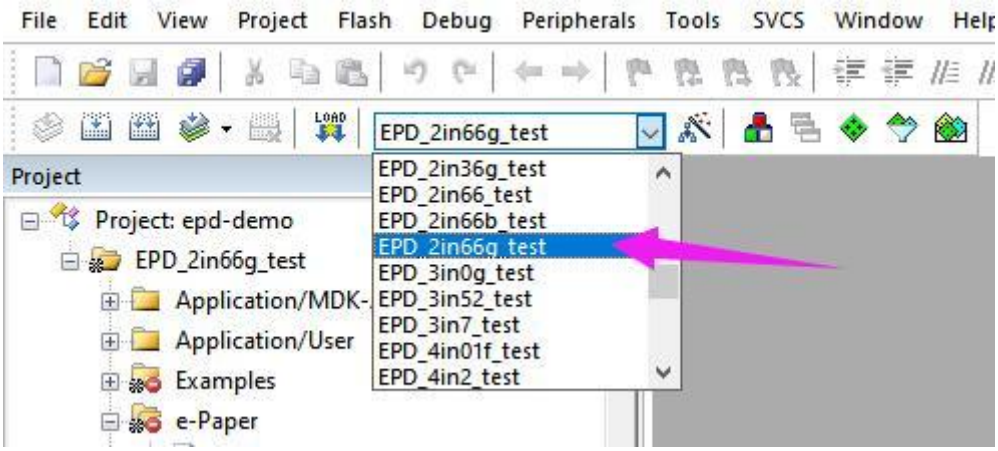
- Click to download the demo, and then unzip it into the "E-Paper_code" directory to see the following content.



- Use Keil to open "epd-demo.uvprojx" in the "E-Paper_code\STM32\STM32-F103ZET6\MDK-ARM" directory
- Open Keil's compilation toolbar (usually already open).



- Select the EPD_2in66g_test at the location shown in the picture.



- Click to compile.

```
linking...
Program Size: Code=24732 RO-data=19288 RW-data=396 ZI-data=53428
FromELF: creating hex file...
"epd-demo\epd-demo.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:12
```

- Make sure the appropriate programmer is connected, then click LOAD to download the demo to the microcontroller.

Resources

Documentation

- User Manual
- Schematic

Demo Codes

- Demo (zip file)
- GitHub

Development Resources

- E-Paper Floyd-Steinberg
- E-Paper API Analysis
- Zimo221.7z
- e-Paper Font Library Tutorial
- Image2Lcd.7z
- Image2Lcd Image Bitmap Conversion
- pwnagotchi Usage

Certification

- CE RoHS

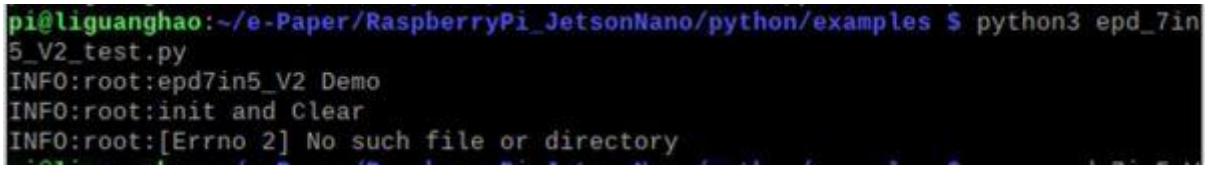
Related Links

- Raspberry Pi Documentation
- e-Paper Dashboard
- Quectel-Pi e-Paper Reader
- GPS Tracker
- TuyaOpen e-Paper Driver

FAQ

Questions about Software

Question: The Raspberry Pi runs the Python demo and the following occurs?



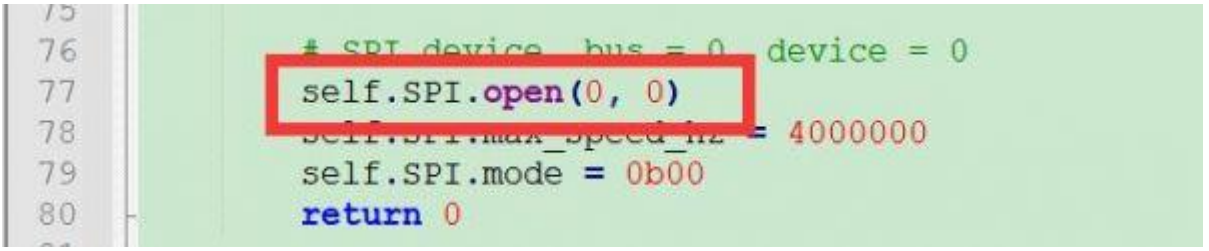
Answer:

- Enter the command: ls /dev/spi*.
- The result may appear as shown in the figure.

```
pi@raspberrypi:~ $ ls /dev/spidev*
/dev/spidev0.1  /dev/spidev1.1  /dev/spidev1.2
```

- This is because the SPI interface is occupied in the /boot/config.txt file.

```
dtparam=spi=on
dtoverlay=spi1-3cs
dtoverlay=mcp251xfd,spi0-0,interrupt=25
dtoverlay=mcp251xfd,spi1-0,interrupt=24
```

- Steps:
  - Delete the occupation of spi0-0 in the /boot/config.txt file.
  - Modify the location shown in the picture in the /e-Paper/RaspberryPi_JetsonNano/lib/waveshare_epd/epdconfig.py file and change it to 0,1.
  - 

Question: How to use custom fonts?

Answer:

- Currently, we do not provide additional font files. You may refer to the custom font generator shared by users to generate and test font files.
- Note: This tool is not officially maintained and is for learning purposes only.
  - https://github.com/ayoy/fontedit

Question: STM32 drives the e-Paper screen, the MDK compilation display space is not enough?

Answer:

- Our demo uses STM32f103ZET6. If the customer modifies other models in MDK, such as STM32F103RBT6, the RAM space becomes smaller, and the stack size and heap size in the startup file need to be modified on the original basis.

Question: Questions about data sheet. When does the font transmit Data Start Transmission 1, and when does it use Data Start Transmission 2?

Answer:

- When transmitting B/W data, use Data Start Transmission 1; When transmitting RED data, use Data Start Transmission 2.

Question: e-Paper shows black border?

Answer:

- The border display color can be set through the Border Waveform Control register or the VCOM AND DATA INTERVAL SETTING register.

Question: After multiple positions are partially refreshed, the font is lighter after refreshing several times?

Answer:

- In this case, the customer needs to reduce the position of the partial refresh and clear the screen after 5 rounds of partial refreshing (increasing the voltage of VCOM can improve the color, but it will increase the afterimage).

Question: When the e-Paper screen is in deep sleep mode, there will be a problem that the screen refresh will not be clean when it wakes up for the first time. How can I solve it?

Answer:

- The process of re-awakening the e-Paper screen is actually the process of re-powering. Therefore, when the EPD wakes up, the screen must be cleared first, so as to avoid the afterimage phenomenon to the greatest extent.

Question: When testing the program, the program has been stuck in e-Paper busy?

Answer:

- It may be caused by the unsuccessful SPI driver.

    - 1. First check whether the wiring is correct.
    - 2. Check whether the SPI is enabled and whether the parameters are configured correctly (SPI baud rate, SPI mode and other parameters).

Question: Why can't the image be displayed when full refreshing after partial refreshing?

Answer:

- The full refresh initialization function needs to be added when the e-Paper screen is switched from partial refresh to full refresh.

Question: Why is the printing information normal when running a Python program, but the e-Paper screen does not respond?

Answer:

- It may be a demo based on the BCM2835 library that has run the C language before. At this time, you need to restart the Raspberry Pi and then run the Python demo.

Question: ImportError: No module named Image?

Answer:

- Install the imaging library using the command "sudo apt-get install python-imaging".

Questions about Hardware

Question: Can Arduino 5V drive the e-Paper?

Answer:

- Yes, now there is a level conversion chip onboard, supporting a 5V drive.

Question: What should be paid attention to when designing the driver board?

Answer:

- The rated input voltage of the e-Paper screen is 2.3~3.6V. If it is a 5V system, level conversion is required. In addition, the voltage should not be lower than 2.5V, so as not to affect the display effect of the e-Paper screen.
- Device selection can use the model in the schematic diagram we provide or choose according to the datasheet.

Question: Can I use analog SPI?

Answer:

- Yes, pay attention to the correct timing.

Question: Why is the BUSY pin always busy?

Answer:

- Check if SPI communication is normal.
- Confirm whether the BUSY pin is normally initialized to input mode.
- It may be that there is no normal reset, try to shorten the duration of the low level during reset (because the power-off switch is added to the drive circuit, the reset low level is too long, which will cause the drive board to power off and cause the reset to fail).
- If the busy function sends the 0x71 command, you can try to comment it out.

Question: What is the specification of the screen cable interface?

Answer:

- 1.64inch, 2.36inch, 3inch, 0.5mm pitch, 26Pin.
- 1.02inch, 0.5mm pitch, 30Pin.
- 4.37inch, 7.3inch, 0.5mm pitch, 50Pin.
- The rest (non-parallel ports) are 0.5mm pitch, 24Pin.

Question: What type of connector does the e-Paper screen use?

Answer:

- Cable socket 0.5-XXpin rear-flip 2.0H (FPC connector).

Questions about Screen

Question: What is the usage environment of the e-Paper screen?

Answer:

- 【Working conditions】Temperature range: 0~40°C (seven-color screen: 15~35°C); Humidity range: 35%~65%RH.
- 【Storage conditions】Temperature range: below 30°C; Humidity range: below 55%RH; Maximum storage time: 6 months.
- 【Transportation conditions】Temperature range: -25~50°C; Maximum transportation time: 10 days.
- 【After unpacking】Temperature range: 20°C±5°C; Humidity range: 50±5%RH; Maximum storage time: Assemble within 72 hours.

Question: Precautions for e-Paper screen refresh?

Answer:

- Refresh mode
    - Full refresh: The e-Paper screen will flicker several times during the refresh process (the number of flickers depends on the refresh time), and the flicker is to remove the afterimage to achieve the best display effect.
    - Partial refresh: The e-Paper screen has no flickering effect during the refresh process. Users who use the partial refreshing function note that after refreshing several times, a **full refresh** operation should be performed to remove the residual image, otherwise the residual image problem will become more and more serious, or even damage the screen (currently only some black and white e-Paper screens support partial refreshing, please refer to product page description).
- Refresh rate
    - During use, it is recommended that customers set the refresh interval of the e-Paper screen to at least 180 seconds (except for products that support the partial refresh function).

- During the standby process (that is, after the refresh operation), it is recommended that the customer set the e-Paper screen to sleep mode, or power off (the power supply part of the e-Paper screen can be disconnected with an analog switch) to reduce power consumption and prolong the life of the e-Paper screen. (**If some e-Paper screens are powered on for a long time, the screen will be damaged beyond repair.**)
- During the use of the three-color e-Paper screen, it is recommended that customers update the display screen at least once every 24 hours (if the screen remains the same screen for a long time, there will be a screen burn that is difficult to repair).

- Usage Environment

  - The e-Paper displays are recommended for indoor use and not for outdoor use.
  - If the usage scenario is outdoors, we do not guarantee the display effect. If the e-Paper screen is damaged due to outdoor use, we do not provide warranty service.
  - Here are some protective measures for outdoor use, but we do not guarantee that the e-Paper screen will function normally even after taking these precautions:
    - Avoid exposing the e-Paper screen to direct sunlight and ensure UV protection. Prolonged exposure to strong light can dry out the charged particles, rendering them inactive and unable to refresh, which is irreversible.
    - Completely cover the white glue part of the e-Paper screen's connection ribbon with 3M tape. Complete coverage and no coverage show different effects under UV light.
    - Place the e-Paper screen in relatively shaded areas, such as under trees or the shadow of eaves.
  - When designing e-Paper screen products, customers should ensure that the usage environment meets the requirements of the e-Paper screen.

Question: What is the refresh rate/lifetime of the e-Paper screen?

Answer:

- Ideally, with normal use, it can be refreshed 1,000,000 times (1 million times).

Question: After using for a while, the screen refresh (full refresh) has a serious afterimage problem that cannot be repaired?

Answer:

- Power on the development board for a long time, after each refresh operation, it is recommended to set the screen to sleep mode or directly power off, otherwise, the screen may burn out when the screen is in a high voltage state for a long time.

Question: After the e-Paper screen enters deep sleep mode, can it be refreshed again?

Answer:

- Yes, but you need to re-initialize the electronic paper with software.

Question: Why is the image displayed offset?

Answer:

- Maybe the SPI rate is too high, resulting in data loss, try to reduce the SPI rate.
- Insufficient or unstable power supply leads to data loss.
- The data cable is so long to cause data loss, the extension cable should not exceed 20cm.

Question: What is the waveform file of the e-Paper screen and what does it do?

Answer:

- The display gray scale of electrophoretic electronic paper is determined by the spatial position of the particles in the Microcapsule or Microcup. The electrophoresis phenomenon occurs between black particles and white particles under the action of voltage. This voltage sequence that promotes the electrophoretic movement of the particles is the driving waveform of the electronic paper. The driving waveform is the core part of the electronic paper display, and the optimization of the driving waveform will directly affect the display effect of the display. The driving waveform file is used to describe the parameters formed by the voltage sequence that promotes the electrophoretic

movement of the particles, and it needs to be called regularly when the electronic paper is refreshed.

- For different batches of e-Paper diaphragms, electrophoretic matrices require different voltage values when driving the display due to materials, manufacturing processes, etc. The waveform of the e-Paper screen is reflected in the relationship between grayscale, voltage and temperature. Generally speaking, after each batch of electrophoresis matrix is generated, there will be a corresponding waveform file in the form of a .wbf file. The film manufacturer will provide the waveform file and electrophoresis matrix to the manufacturer of the e-Paper screen, and then the manufacturer of the e-Paper screen integrates the protection board, substrate and driver and then provides it to customers; if the waveform file does not correspond to the screen, it is likely that the display cannot be displayed or the display effect is unsatisfactory. Generally, the waveform file has OTP built into the driver IC of the e-Paper screen when leaving the factory, and some programs we provide also called external waveform files to drive the e-Paper screen.

Question: What do LUT and OTP stand for?

Answer:

- LUT is the abbreviation of LOOK UP TABLE, and OTP is the abbreviation of ONE TIME PROGRAM. The original intention of LUT is to load waveform files, and the waveform files are divided into OTP and REGISTER. Among them, OTP is the built-in waveform storage method, and REGISTER is the external waveform storage method.

Question: What is the process of partial refreshing?

Answer:

- There are mainly two types of e-Paper screens.
    - One is to refresh the background image first.
    - The other is to alternately refresh old data and new data.

Question: How to partially refresh in different positions at the same time?

Answer:

- Simultaneous partial refreshing in different locations needs to be operated in the program design, that is, first refreshing the data of different locations into the electronic paper IC, and finally doing the Update/TurnOnDisplay uniformly.

Question: Does the three-color e-Paper have a red/yellow color difference?

Answer:

- Yes, when the e-Paper is batched, there will be some color difference, which is a normal phenomenon. Store the e-Paper faces up to reduce the reddish/yellowishness to a certain extent.

Question: Are bare screens shipped with a film?

Answer:

- With film.

Question: Does e-Paper have a built-in temperature sensor?

Answer:

- At present, all screens have built-in temperature sensors, and you can also use IIC pins external LM75 temperature sensor.